

Abstract Algebra — Homework 2

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(a)

By the tower law, $[K : \mathbb{R}] = [K : \mathbb{C}][\mathbb{C} : \mathbb{R}] = 2^{a+1}m$. Let L be the Galois closure of K over $\mathbb{R}$. Set $G = \text{Gal}(L/\mathbb{R})$, so $|G| = [L : \mathbb{R}]$. It's obvious that $|G| = 2^b m'$ for some $b \leq a, m' | m, m'$ is odd (also by tower law).

Let $P \leq G$ be a sylow 2-subgroup of G , so $[G : P] = m'$ is odd. Let F be the corresponding intermediate field, we have $[F : \mathbb{R}] = [G : P] = m'$. Suppose $\alpha \in F \setminus \mathbb{R}$, and its minimal irreducible polynomial has degree d ($d \geq 2$). Clearly, $d | [F : \mathbb{R}]$ so d is odd. But any odd-degree polynomial has a real root, which contradicts the irreducibility. Hence, $F = \mathbb{R}$ and $G = P$ is a 2-group.

Therefore, $[L : \mathbb{R}] = 2^b$, by the tower law, $[L : \mathbb{R}] = [L : \mathbb{C}][\mathbb{C} : \mathbb{R}] = 2[L : \mathbb{C}]$, $[L : \mathbb{C}] = 2^{b-1}$. Since $[K : \mathbb{C}]$ divides $[L : \mathbb{C}]$, the only possibility is $a = 0$.

(b)

Assume, $a \leq 1$. Now define $H = \text{Gal}(L/\mathbb{C}) \leq G$, the subgroup is not trivial since $|H| = [L : \mathbb{C}] = [L : K][K : \mathbb{C}] \leq 2^a \leq 2$

Since 2 is the least prime dividing $|H|$, there exists a normal subgroup of H of index 2, let it be N , $[H : N] = 2$. By the Galois correspondance, the fixed field $E = L^N$ satisfies $[E : \mathbb{C}] = [H : N] = 2$, so E is a degree-2 extension of $\mathbb{C}$.

Pick any $\beta \in E \setminus \mathbb{C}$, its minimal irreducible polynomial over $\mathbb{C}$ must be quadratic:

$$x^2 + bx + c, b, c \in \mathbb{C}$$

By the formula, the roots are $\frac{-b \pm \sqrt{b^2 - 4ac}}{2}$, now $b^2 - 4ac \in \mathbb{C}$ and every complex number has its square root in $\mathbb{C}$ trivially, so $\beta \in \mathbb{C}$, a contradiction.

Hence, $a = 0$.

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(a)

The minimal polynomial of $\sqrt[3]{2}$ over $\mathbb{Q}$ is $x^3 - 2 = (x - \sqrt[3]{2})(x^2 + 2x + 2)$, which does not split in $\mathbb{Q}(\sqrt[3]{2})$. Hence, $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not a normal extension.

(b)

Lemma 2.1. *Splitting Fields are Normal Extensions.*

Proof. If $f(x) \in K[x]$ and L is the splitting field of $f(x)$ over K , then we can directly assume $L = K(a_1, \dots, a_n)$, where $a_1, \dots, a_n \in L$ are all the roots of $f(x)$. Since adjoining finitely many algebraic roots generates a finite-dimensional vector space over the base field, we know that L/K is a finite extension. Next, we will prove that L/K is a normal extension. If $\alpha \in L$ and its minimal polynomial over K is $p(x)$, we need to show that all roots of $p(x)$ lie in L .

Let β be another root of $p(x)$. Since $p(x)$ is an irreducible polynomial in $K[x]$ and α, β are its roots, we know there exists a K -isomorphism $\phi : K(\alpha) \rightarrow K(\beta)$. Since $f(x) \in K[x]$, we have $f(x) \in K(\alpha)[x]$. Furthermore, since ϕ fixes the elements of K , we obtain $f^\phi(x) = f(x)$. Now, considering $f(x)$ as a polynomial in $K(\alpha)[x]$, we know $K(\alpha)(a_1, \dots, a_n)$ is the splitting field of $f(x)$ over $K(\alpha)$. On the other hand, considering $f^\phi(x) = f(x)$ as a polynomial in $K(\beta)[x]$, we know $K(\beta)(a_1, \dots, a_n)$ is the splitting field of $f^\phi(x)$ over $K(\beta)$.

Because $K(\alpha)(a_1, \dots, a_n) = K(a_1, \dots, a_n)(\alpha) = L(\alpha)$ and $\alpha \in L$, we have $K(\alpha)(a_1, \dots, a_n) = L$. Similarly, we have $K(\beta)(a_1, \dots, a_n) = K(a_1, \dots, a_n)(\beta) = L(\beta)$. Rearranging these results: we have an isomorphism $\phi : K(\alpha) \rightarrow K(\beta)$ and a polynomial $f(x) \in K(\alpha)[x]$, and additionally, L and $L(\beta)$ are the splitting fields of $f(x)$ and $f^\phi(x)$ over $K(\alpha)$ and $K(\beta)$, respectively. Because an isomorphism between base fields can always be extended to an isomorphism between their corresponding splitting fields, we know there exists an isomorphism $\sigma : L \rightarrow L(\beta)$ such that $\sigma|_{K(\alpha)} = \phi$.

Since ϕ fixes the elements of K , σ also fixes the elements of K . That is to say, σ is a K -isomorphism, meaning L and $L(\beta)$ are isomorphic over K . Since isomorphic field extensions must have the same degree over their base field, we know $[L : K] = [L(\beta) : K]$. Thus, from the multiplicativity of degrees in towers of extensions, $[L(\beta) : K] = [L(\beta) : L][L : K]$, we obtain $[L(\beta) : L] = 1$. This implies $L(\beta) = L$, which means $\beta \in L$. We have proved that all roots of $p(x)$ must be in L , meaning that $p(x)$ splits over L ; hence, by definition, L/K is a normal extension. $\square$

Since $\mathbb{Q}(\sqrt[4]{2}, \zeta_8)/\mathbb{Q}$ contains all roots of $x^4 - 2$ ($\sqrt[4]{2}, -\sqrt[4]{2}, \zeta_8^2 \sqrt[4]{2}, -\zeta_8^2 \sqrt[4]{2}$), it's a splitting field for the irreducible polynomial (in $\mathbb{Q}[x]$) $x^4 - 2$. Every splitting field is normal, so $\mathbb{Q}(\sqrt[4]{2}, \zeta_8)/\mathbb{Q}$ is normal.

Clearly, $\mathbb{Q}(\sqrt[4]{2}, \zeta_8) = \mathbb{Q}(\sqrt[4]{2}, i)$. By tower law, $[\mathbb{Q}(\sqrt[4]{2}, i) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[4]{2}, i) : \mathbb{Q}(i)][\mathbb{Q}(i) : \mathbb{Q}] = 4 \times 2 = 8$. Hence, the degree for the extension is 8.

3

Let E/K be an arbitrary finite extension. If $E = K$, then $[E : K] = 1 = p^0$, so we may assume that $E \neq K$.

Because K has characteristic 0, every algebraic extension of K is separable, therefore, E/K is a finite separable extension.

In field theory, the **primitive element theorem** states that every finite separable field extension is simple, i.e. generated by a single element. So, there exists a Galois closure N of E over K . (Since we can write $E = K(\alpha)$ for some α as a finite extension over K , and α is algebraic over K , there exists a unique irreducible polynomial f over K , and then by adjoining all roots of f , we can construct the splitting field N , which is normal and separable.)

Let $G = \text{Gal}(N/K)$ be the Galois group, by definition $|G| = [N : K] = p^a m$ for some positive integer a and m ($p \nmid m$). Assuming the case m is greater than 1. By Sylow's first theorem, there exists a Sylow p -subgroup P with order $|P| = p^a$. By the fundamental theorem of Galois theory, there's a corresponding intermediate fixed field $L = N^P$ such that $K \subseteq L \subseteq N$.

We have $[L : K] = [G : P] = \frac{|G|}{|P|} = \frac{p^a m}{p^a} = m$, since we assume $m > 1, p \nmid m$ there's a contradiction.

Hence, $|G| = [N : K] = p^a$. By the tower law, $[N : K] = [N : E][E : K]$ and $[E : K] > 1$, $[E : K]$ must be some power of p , completes the proof.

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Let $G = L/K$ be the Galois group, by definition $|G| = [L : K] = p^m$.

We claim that there exists a sequence of subgroups $G = H_0 \supsetneq H_1 \supsetneq \dots \supsetneq H_m = \{e\}$ such that for every $1 \leq i \leq m$:

1. $H_i \triangleleft G$
2. $[H_{i-1} : H_i] = p$

Let's prove the claim by induction. For the base case ($m = 1$), $|G| = p$, by Lagrange theorem H_m must be trivial, and the two statements are satisfied.

Assume the lemma holds for all p -group of order p^{m-1} ($m > 1$), because G is non-trivial p -group, by class equation:

$$|G| = |Z(G)| + \sum [G : C_G(x_i)]$$

where x_i 's belong to distinct conjugacy classes and not in $Z(G)$, since the terms within the sum are all some power of p , so is $|G|$, we know $p \mid |Z(G)|$, the center of G is not trivial.

By Cauchy's theorem, $Z(G)$ contains an element of order p , which can generate a cyclic group N of order p , and obviously $N \triangleleft G$.

Consider the quotient group G/N , which has order p^{m-1} . By our inductive hypothesis, G/N contains a chain of normal subgroups:

$$G/N = \overline{H_0} \supsetneq \overline{H_1} \supsetneq \dots \supsetneq \overline{H_{m-1}} = \{e\}$$

such that each $\overline{H_i} \triangleleft G/N$ and $[\overline{H_{i-1}} : \overline{H_i}] = p$.

By the fourth isomorphism theorem (correspondance theorem), each quotient corresponds uniquely to a subgroup of H_i of G containing N . Because $\overline{H_i}$ is normal in G/N , its corresponding preimage H_i is normal in G . Furthermore, the index is preserved, $[H_{i-1} : H_i] = [\overline{H_{i-1}} : \overline{H_i}]$. The mapping details can be seen [here](#). By some observation, we know $H_{m-1} = N$, set $H_m = e_G$, we have $[H_{m-1} : H_m] = p$, completes our induction. **The claim is proved so perfectly!**

Now define K_i to be the fixed field of H_i , by the Galois correspondance we have the chain:

$$K = K_0 \subsetneq K_1 \subsetneq \dots K_m = L$$

K_i/K is Galois if and only if $\text{Gal}(K_i/K) \triangleleft G$, since our lemma explicitly construct each $H_i \triangleleft G$, so the second statement is True.

Using tower law for the extensions $K_{i-1} \subsetneq K_i \subseteq L$, we have:

$$[L : K_{i-1}] = [L : K_i][K_i : K_{i-1}]$$

So, $|H_{i-1}| = |H_i|[K_i : K_{i-1}]$. We have $[K_i : K_{i-1}] = [H_{i-1} : H_i] = p$, completing the proof.